

Forward and Inverse Problems in Elasticity Imaging of Soft Tissues

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Abstract

In elasticity imaging, a surface deformation is applied to an object using small pistons, and the resulting induced strains in the interior of the object are measured using ultrasonic imaging. Two important problems are considered: (1) the forward problem of determining the strains induced by a known deformation of an object with known elasticity; and (2) the inverse problem of reconstructing elasticity from measured strains and the equations of equilibrium. The method of finite differences is used to solve the forward problem for a given piston configuration; some non-trivial issues arise in determining boundary conditions. The finite difference equations are then rearranged into a linear system of equations which formulates the inverse problem; this system can be solved for the unknown elasticities. This formulation of the inverse problem is completely consistent with the forward problem; this is useful for iterative methods in which the deformation is adaptively changed. A comparison between simulated and actual measured results demonstrate the feasibility of the proposed procedure.

I. INTRODUCTION

An important problem in medicine is the detection of tumors and lesions in the body. Even today, despite great advances in developing new imaging modalities, a simple and reliable technique to detect cancers has not yet been developed. Palpation is still widely used to detect hard masses in the testes and breast as a self-screening process. However, its reliability is restricted to lesions relatively close to the periphery - the method cannot reliably be used to detect tumors buried deeper in the body.

Recently, elasticity imaging has been proposed by many researchers as a new diagnostic tool for several pathologies, such as breast cancer [1, 2, 3, 4, 5, 6, 7, 8, 9]. The goal

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of elasticity imaging is to develop alternative methods of palpation, in order to expand palpation's limited range to include deep-lying lesions. Such imaging may permit differential diagnosis based on the order-of-magnitude difference that exists between the Young's moduli (elasticities) of normal and cancerous tissue. However, this cannot be done easily, since no current imaging modality provides direct information about this fundamental tissue parameter.

Several authors have recognized the potential of using the elastic properties of tissue as a means of differentiating between cancerous and normal tissue. Parker *et al* [6] demonstrate the effect of applying low-frequency mechanical vibrations to a phantom; they use a finite-element model to measure the modulus of elasticity using the appropriate mathematical description of the object. Tristram *et al* [7, 8] quantify tissue motion *in vivo* from the correlation coefficient between pairs of ultrasound scans. They also propose the use of Fourier-based techniques to study patterns of tissue movement. Bertrand *et al* [10] suggest a speckle tracking algorithm to assess the contractive performance of muscles. Krouskop *et al* [11] studied the mechanical properties of soft tissue to predict the interaction of a prosthetic socket with an amputee's residual limb. Bamber *et al* [3] use a semi-quantitative measure to evaluate the real-time features of the breast. They use feature extraction and pattern recognition techniques to classify the usefulness of each feature in diagnosis.

Several other authors [9, 12, 13] have proposed imaging systems that can be used to measure tissue motion. O'Donnell *et al* [4] have developed Fourier-based algorithms for tracking arterial wall motion that demonstrates tissue motion variation in the presence of harder plaques. The spatial variation of tissue elastic properties has also been evaluated by Adler *et al* [1, 2], who used cardiac motion as a source of vibration to predict fetal lung maturity.

However, a fundamental problem with these approaches has been the assumption of the known geometry and location of lesions in the part of the body under consideration. In such cases, it is clear how to deform the body in order to obtain the "best" strain map, i.e., the map which provides the clearest image of the elasticity. But this information is not always available. Indeed, lesions can have an arbitrary geometry and be located anywhere within the body.

To deal with this, a more rigorous approach that does not require any knowledge about the presence or location of the inclusion is needed. One approach to elasticity imaging is to reconstruct mechanical properties of tissue based on the measurement of initial motion under the action of externally applied forces. Yemelyanov *et al* [9] proposed a speckle-tracking algorithm to measure tissue motion, to detect a hard inclusion with a known geometry at the center of a cylindrical phantom.

Our goal is to translate the intuitive notions of palpation into a quantitative form that is amenable to mathematical manipulation. To achieve this, the 2-D partial differential equations of equilibrium are derived assuming an incompressible medium. These equations are solved for the unknown functions, using the method of finite differences (FD). Our approach differs from previous work in that no assumption is made about the presence or locations of inclusions. It is related to the work being done by O'Donnell *et al* [4], Yemelyanov *et al* [9], and Skovoroda *et al* [14], [15]. However, our work differs from that of [14] and [15] in that we solve the inverse problem (IP) by rearranging the equations of the forward problem (FP), thus preserving consistency between the two problems. Also, we solve both the forward and inverse problems using LU decomposition, which is much faster than the iterative methods used in [14] and [15]. For more details concerning the actual measurement of strains we refer the reader to [15].

Two problems of interest are identified here. In the FP, the elasticity of the object is known and we solve for the displacements (strains). In the IP, the measured displacements (strains) are used to solve for the unknown elasticity of the object.

The paper is organized as follows. In Section II, the problem is formulated and the equations of equilibrium are presented. This is followed by a discussion of the FP in Section III. A comparison between our simulation results and the experimental data presented in [15] shows good correlation, validating our FD equations. In Section IV, the IP is solved by rearranging these FD equations into a linear system of equations with the measured strains as the known quantities and the elasticity as unknown. Note that this system of equations is consistent with the forward problem FD equations which have been experimentally validated. Simulations, including noisy data, show that the elasticities may be reconstructed. Finally, in Section V concluding remarks and future research endeavors, including an adaptive surface deformation procedure in which the surface deformation which best exhibits an approximately-reconstructed inclusion is determined, are outlined.

II. EQUILIBRIUM EQUATIONS

A. Basic Equations

Let τ_{ij} represent the components of stress in direction j acting on a planar surface with normal direction i , where

$1 \leq i, j \leq 3$ for 3-dimensional space. Let the normal to the plane be $\mathbf{n}$, with components $\mathbf{n}_1$, $\mathbf{n}_2$ and $\mathbf{n}_3$ in directions 1, 2, 3, respectively. Then the stress $\mathbf{T}_i$ acting on the body along $\mathbf{n}$ is given by

$$\mathbf{T}_i^n = \tau_{i1}\mathbf{n}_1 + \tau_{i2}\mathbf{n}_2 + \tau_{i3}\mathbf{n}_3. \quad (1)$$

Next, let X_i represent the components of the external body force per unit volume acting on the body. For a body in equilibrium, the following differential equation must be satisfied:

$$\frac{\partial \tau_{i1}}{\partial x_1} + \frac{\partial \tau_{i2}}{\partial x_2} + \frac{\partial \tau_{i3}}{\partial x_3} + X_i = 0, \quad i = 1, 2, 3. \quad (2)$$

We assume that the object under consideration is a perfectly elastic solid. For such a material, the relation between stress and strain is described by Hooke's law:

$$\sigma_{ij} = C_{ijkl}e_{kl}, \quad (3)$$

where σ_{ij} is the stress tensor, e_{kl} is the strain tensor, and C_{ijkl} is a tensor of elastic constants that are independent of stress or strain. In (3) summations over k and l are not indicated explicitly. For isotropic media (3) simplifies to

$$\begin{aligned} \sigma_{11} &= \lambda(e_{11} + e_{22} + e_{33}) + 2Ge_{11} \\ \sigma_{22} &= \lambda(e_{11} + e_{22} + e_{33}) + 2Ge_{22} \\ \sigma_{33} &= \lambda(e_{11} + e_{22} + e_{33}) + 2Ge_{33} \\ \sigma_{12} &= 2Ge_{12} \\ \sigma_{23} &= 2Ge_{23} \\ \sigma_{31} &= 2Ge_{31}, \end{aligned} \quad (4)$$

where λ and G (or μ) are the Lamé constants. These equations can be solved for e_{ij} ; however, the inverse form is usually written as

$$\begin{aligned} e_{11} &= \frac{1}{E} [\sigma_{11} - \nu(\sigma_{22} + \sigma_{33})]; \\ e_{12} &= \frac{1 + \nu}{E} \sigma_{12} = \frac{1}{2G} \sigma_{12}. \end{aligned} \quad (5)$$

Similar expressions can be written for the other quantities as well. The constants E and ν are related to the Lamé constants by

$$\begin{aligned} \lambda &= \frac{2G\nu}{1 - 2\nu} = \frac{E\nu}{(1 + \nu)(1 - 2\nu)}; \\ G &= \frac{E}{2(1 + \nu)}, \end{aligned} \quad (6)$$

where E is the Young's Modulus, ν is Poisson's ratio, and G is the shear modulus of the material. For incompressible media, $\nu = \frac{1}{2}$ and λ converges to a (unknown) pressure term, which must be added to (4).

Let $E(x, y) = E_0\mathcal{E}(x, y)$ represent the Young's modulus of elasticity of the object; and let $u(x, y)$ and $v(x, y)$ be the distribution of the horizontal and vertical components of displacement in the object under the influence of the boundary deformation(s). Since the object is assumed to be incompressible, an unknown pressure term $p(x, y)$ enters the equilibrium equations [16].

B. Equilibrium Equations

The 2-D equations of equilibrium for the object can be shown to be ¹[15]

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + 2\mathcal{E}_1 \frac{\partial u}{\partial x} + \mathcal{E}_2 \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) + \lambda \frac{\partial p}{\partial x} = 0 \quad (7)$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + 2\mathcal{E}_2 \frac{\partial v}{\partial y} + \mathcal{E}_1 \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) + \lambda \frac{\partial p}{\partial y} = 0 \quad (8)$$

where $\mathcal{E}_1 = \lambda \frac{\partial \mathcal{E}}{\partial x}$, $\mathcal{E}_2 = \lambda \frac{\partial \mathcal{E}}{\partial y}$ and $\lambda = \frac{1}{\mathcal{E}}$. The condition of incompressibility leads to

$$\vec{\nabla} \cdot \vec{d} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad (9)$$

where $\vec{d}$ is the total displacement in the object interior.

C. Finite Differences

We solve the PDEs (7)-(9) using finite differences (FD) to discretize the region over which the PDE is defined using a finite grid. An index is assigned to each grid point and the PDE is evaluated at each grid point using suitable discrete approximations to the derivatives. This results in a system of simultaneous linear equations, which can be written in matrix form as $Ax = b$.

For the case of a single function, Fig. 1 shows a possible indexing scheme. Each point in the grid is assigned a number, with negative indices representing points outside the region of interest. The boundary of the body is extended in order to use more accurate approximations to the derivatives at the boundary. The negative indices can be resolved in terms of the indices within the body by discretizing the boundary conditions; this yields a relation between points outside the body and points within the body.

This technique can be extended in a straightforward manner to the case where multiple PDEs are used to describe multiple unknown functions. For N functions, in order to uniquely solve the system, we need N PDEs. The PDEs are discretized over an $M \times M$ grid and the equations written out for each grid point are stacked together to produce a composite system which can be written in matrix form as $A\mathbf{x} = \mathbf{b}$, where A is an $M^2N \times M^2N$ (or smaller) matrix of coefficients and $\mathbf{x}$ is the composite unknown vector of dimension $(M^2N) \times 1$ given by

$$\mathbf{x} = [f_1^T \ f_2^T \ \dots \ f_N^T]^T; \quad (10)$$

$$f_i^T = [f_1^i \ f_2^i \ \dots \ f_M^i]^T,$$

where f_j^i is the j^{th} unknown component at the i^{th} grid point of the unknown function, following the indexing scheme of Fig. 1 for each component.

In discretizing the PDEs for our problem, instead of extending the boundary beyond the physical boundaries of

¹For convenience, dependence on (x, y) is dropped where no confusion can occur.

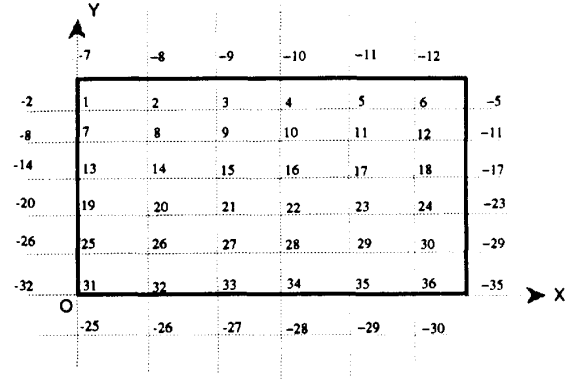


Figure 1: Indexing for a single function. The dark outline is the region of interest while the light region represents the extension of the boundaries to compute derivatives at the boundary

the object, we use higher order difference approximations using points only within the object, e.g., we use a three-point $O(h^2)$ difference formula to approximate the first derivative at the boundaries.

It is relevant at this point to note that the equations of equilibrium can be derived for the general 3-D case [14]. Although numerical solution is theoretically possible, current ultrasonic equipment can only produce planar (2-D) information. In [14], Skovoroda *et al* show that unlike a 1-D model, reconstructions based on a 2-D analysis are not in great error, even if a plane deformed state is not maintained, as long as a uniform external surface deformation is applied.

III. THE FORWARD PROBLEM

For the forward problem, the elasticity of the object is known and equations (7)-(9) must be solved for the unknown displacements (strains). Since the pressure is also unknown, we have 3 PDEs and 3 unknown functions.

A. Finite Difference Discretization

Discretization of (8) can be performed by using central differences for the first derivatives of the displacement and the five-point formula for the Laplacian. In using the FD approximation formulae, we ensure that at least $O(h^2)$ approximations are used for all derivatives. Performing the discretization at the point (i, j) in the grid results in

$$\begin{aligned} & \left(1 + \frac{\mathcal{E}_1 h}{2}\right) v_{i+1,j} + (1 + \mathcal{E}_2 h) v_{i,j+1} + \\ & \left(1 - \frac{\mathcal{E}_1 h}{2}\right) v_{i-1,j} + (1 - \mathcal{E}_2 h) v_{i,j-1} - 4v_{i,j} + \end{aligned} \quad (11)$$

$$\frac{\mathcal{E}_2 h}{2}(u_{i,j+1} - u_{i,j-1}) + \lambda h(p_{i,j+1} - p_{i,j-1}) = 0.$$

These can be expressed in a compact form as [17]

$$\begin{bmatrix} 1 - \frac{a}{2} & 1 + b & 1 + \frac{a}{2} \\ -4 & 1 - b & 1 + \frac{a}{2} \end{bmatrix} v + \begin{bmatrix} \frac{b}{2} \\ -\frac{b}{2} \end{bmatrix} u + \begin{bmatrix} c \\ -c \end{bmatrix} p = 0, \quad (12)$$

where $a = \mathcal{E}_1 h$, $b = \mathcal{E}_2 h$, $c = \lambda h$, $x = ih$, $y = jh$ and h is the step size, assumed equal in both directions. Adopting a similar procedure for (7) yields

$$\begin{bmatrix} 1 - a & 1 + \frac{b}{2} \\ -4 & 1 + a \\ 1 - \frac{b}{2} & 1 + a \end{bmatrix} u + \begin{bmatrix} -\frac{b}{2} & \frac{b}{2} \end{bmatrix} v + \begin{bmatrix} -c & c \end{bmatrix} p = 0. \quad (13)$$

The equations are generated as follows. First equation (12) is computed over all the unknown v grid points, followed by the computation of equation (13) over all the unknown u points on the u grid. Then the incompressibility equation (9) and the boundary conditions are used to obtain the equations for all the unknown points on the p grid. This yields a matrix equation of the form $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{x} = [\mathbf{v} \ \mathbf{u} \ \mathbf{p}]^T$, where $\mathbf{v}$ is the vector of all unknown v points arranged in the order of increasing index. The $\mathbf{u}$ and the $\mathbf{p}$ vectors are defined similarly.

B. Boundary Conditions for Forward Problem

The boundary conditions used to solve the equations are extremely important. At the (pressure-) free surface, by definition, we have $\sigma_{11} = \sigma_{22} = 0$. We assume that no slippage occurs under the pistons; note that the table itself acts as a piston (see Fig. 2). Without loss of generality, displacements are measured with respect to the lower left corner of Fig. 2. Hence $u = 0$ at the bottom of the object, $v = 0$ under a horizontal piston and $u = 0$ under a vertical piston. We also assume that no cavities develop on the bottom of the object as deformations are applied, so that $v = 0$ at the bottom of the object.

Since it is known that

$$\begin{aligned} \sigma_{11} &= E_0 m(p + 2\mathcal{E}\epsilon_{11}) \\ \sigma_{22} &= E_0 m(p + 2\mathcal{E}\epsilon_{22}) \\ \sigma_{12} &= 2E_0 m\mathcal{E}\epsilon_{12} \end{aligned} \quad (14)$$

where $\epsilon_{11} = \frac{\partial u}{\partial x}$, $\epsilon_{22} = \frac{\partial v}{\partial y}$, $\epsilon_{12} = \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x})$ and $m = \frac{1}{3}$ for an incompressible medium, we get additional constraints at the boundary on the unknown displacements and the pressure. Here and in Fig. 2, we replace the coordinates $i = 1, 2, 3$ with x, y, z , respectively, for clarity.

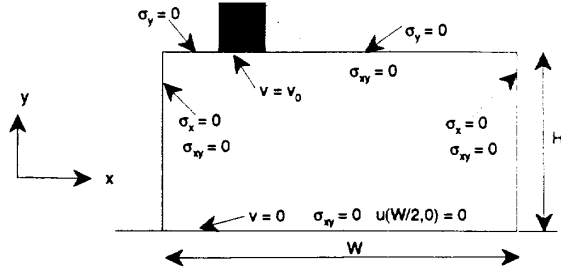


Figure 2: Boundary conditions for the Forward Problem

Parameter	Value
Object width (W)	58.5
Object height (H)	58.5
Grid Size	39 × 39
Homogeneous elasticity	2.0
Hard inclusion elasticity	4.0
Soft inclusion elasticity	1.0
Inclusion geometry	Rectangle, Centered
Boundary deformation/piston	3.0 (5%)

Table 1: Simulation Parameters used to solve the FP

C. Simulations for the Forward Problem

Table 1 shows the simulation parameters used to solve the forward problem. Several different piston configurations were tried, and results are presented for the particular cases of: (1) 2 horizontal pistons; and (2) 1 vertical piston centered on the object. The inclusions were rectangular and symmetrically located in the object. The results are presented for a 39 × 39 grid; the inclusion extended from grid line 11 to 29 in both directions. The boundary deformation was a 5% strain for each piston, and the inclusion magnitudes were twice as large as the homogeneous case for the hard inclusion, and one-half the homogeneous magnitude for the soft inclusion. The equations were generated as described above using an efficient sparse matrix data structure [18], and a direct LU decomposition method was used to solve the linear system of equations.

Fig. 4 shows a plot of the horizontal center grid line for the case of two horizontal pistons centered on the two sides of the object. We expect this plot to be completely symmetrical about a vertical line at the center of the object, since the boundary conditions on either side of this grid line are identical. Three cases of homogeneous, hard and soft inclusion were considered. Note the change in slope relative to the homogeneous case for the hard and soft inclusion grid lines at grid points 11 through 29. This coincides with the boundary of the inclusion. The dip in the hard inclusion plot is expected since smaller strains are induced in harder objects; this also explains the higher strains seen in the soft inclusion case at the inclusion site.

The plot in Fig. 6 shows the vertical center grid line for

the case of deformation by a vertical piston centered on top of the object. Unlike the horizontal piston case, for the case of a vertical piston, there is an asymmetry in the plot which is due to the boundary condition at the bottom being different from that at the top. While the object is constrained to be motionless at the bottom, the boundary at the top is a free surface. This leads to an asymmetry in the plot with larger (negative) strains at the top of the object and smaller strains at the bottom.

The strains for the vertical piston case are negative, since for this case the object is being compressed in the vertical direction. In contrast, for the horizontal piston case the object is being compressed horizontally and hence (because of incompressibility) expanded vertically. In other words, negative strains imply compression in that direction and positive strains represent expansion in that direction.

D. Comparison with Experimental Results

We now compare our simulation results to experimental results from [15]. Fig. 5 shows the results of actual strain measurements taken with one vertical piston centered over a circular phantom [15]. There are several differences between our simulation and this experiment:

- (1) the experimental phantom is circular, while the simulation assumes a rectangular object;
- (2) the experiment used a single vertical piston (opposite the table) with $\approx 6\%$ strain deformation, while the simulation used two horizontal pistons each having a 5% strain (for a total of 10%); and
- (3) the elasticity ratio of the inclusion relative to the surrounding material was 2.5 for the experiment and 2.0 for the simulations.

Despite these differences, comparing Fig. 4 and Fig. 5 we see that the simulation has captured the essential features of the FP. Note that comparing horizontal and vertical displacements is appropriate, since the simulation used horizontal deformations, while the experiment used vertical deformations. The roughly factor of two difference between the resulting strains in Fig. 4 and Fig. 5 is due to the roughly factor of two difference in the applied deformation.

It must be noted in particular that the FP and IP are not trivial; the form of even a gross inclusion cannot easily be discerned from the induced strains alone.

IV. THE INVERSE PROBLEM

We now consider the inverse problem (IP) of reconstructing the elasticities from the displacements or strains. There are two aspects to this problem. The first is to demonstrate that a solution to the IP exists and can be found. The second is to demonstrate the robustness and sensitivity of the solution to the presence of measurement noise. Here we show that the IP can be solved by reconstructing the

elasticities from the displacements obtained in the solution to the FP. We also consider the case of measurement noise by using an appropriate noise model and perturbing the FP solution. From our simulations, it appears that the IP is robust and reconstruction is possible even in the presence of noise (for small grid sizes).

A. Rearrangement of FP Equations

As discussed above, the FD equations for the FP lead to a system of equations of the form $Ax = b$, where x is a vector of the unknown displacements and pressure at the grid points. From the discretized equations (12) and (13) it is clear that these equations can be rearranged with the elasticities and pressure as the unknowns. This yields a different linear system of equations $A'x' = b'$ for the inverse problem. The vector x' now contains the unknown elasticity and pressure at each grid point. Note that in contrast to the FP where we had three unknown functions, the IP has only two unknown functions. The equations (7) and (8) are the two PDEs used to solve for the two unknown functions. The elements of the matrix A' are now functions of the displacements and its derivatives, and the vector b' contains the boundary conditions for the IP which are conditions on the elasticity (assumed known at boundary) and the pressure.

The pressure boundary conditions are the same as those imposed in the FP and are obtained from (14). The boundary conditions and piston configurations were kept identical for the FP and the corresponding IP. It is apparent that the measured data may not satisfy the condition of incompressibility due to measurement errors. Indeed, it is a property of the measuring system that the strain can be measured accurately only in a single direction; i.e., a good estimate of the lateral strain component is obtained at the cost of a poor estimate of the transverse component and vice-versa [15].

One way to offset this would be [14] to estimate the lateral displacement from a good measurement of the vertical displacement using the incompressibility condition (9) and assuming that $u = 0$ along the central vertical axis of the image (for deformation using a centered vertical piston at the top of the object.). This is a very good approximation for the experimental setup of [14], but it may not hold in general clinical situations. However, in [14] it is claimed that by careful control of the deformation, lateral displacements over at least one line of the image can be minimized. Further studies to investigate this are being undertaken.

B. Properties of the Inverse Problem

An interesting issue that arises in solving the IP is what the "optimum" piston configuration should be in order to reconstruct the elasticities. Since the matrix A' tends to be ill-conditioned for fine grids, the question of optimality (possibly non-unique) of the piston configuration may be

posed by using the condition number (CN) of the matrix A' as a criterion. This problem will be examined in another paper.

This trade-off between grid size and CN is typical of FD solutions to PDEs. A fine grid provides better approximations to the solution, but this is attained at the cost of increasing the ill-conditioning of the system matrix and the dimension of the problem (the size of the matrix $\approx 2(N-2)^2$, where N is the grid size). In our simulations, we have found that using a coarse grid for the IP works best—the lower CN is more significant than the better approximation to the solution, thus enabling us to use fast direct methods (LU) to solve the system of equations. If, on the other hand, a fine grid is necessitated by the application, more complex regularized least-squares techniques (e.g., truncated singular value decomposition) must be used to reconstruct the elasticities. These approaches are still being investigated.

In solving the IP in the manner proposed above, several important properties become apparent. First, the solution to the IP obtained above exactly solves the forward problem, since they are derived from the same equations. This consistency is an important feature of the proposed technique. By studying the IP directly, rather than just iterating over the FP to solve the IP, important issues such as the inherent ill-conditioning of the problem and variation of the conditioning of the problem with grid size, can be observed and addressed directly. This cannot be done if iteration on the FP is used.

C. Simulations for the Inverse Problem

Parameter	Value
Object width (W)	58.5
Object height (H)	58.5
Grid Size	15 × 15
Homogeneous elasticity	2.0
Hard inclusion elasticity	4.0
Soft inclusion elasticity	1.0
Inclusion geometry	Rectangle, Centered
Boundary deformation/piston	3.0 (5%)

Table 2: Simulation Parameters used to solve the IP

Table 2 shows the simulation parameters used to solve the inverse problem. Two different piston configurations are used to show the dependence of CN on piston configuration. For the example considered, both piston configurations yielded a well-conditioned IP matrix A' , and the elasticities are successfully reconstructed for both cases. This is shown in Fig. 7, where the reconstructed elasticity for the case of a vertical push is plotted for the case of a large hard inclusion in the middle of the object. Fig. 9 shows the case of soft inclusion with external deformation using 2 horizontal pistons centered on the sides of the object. The reciprocal of the condition number (CN) of the

matrix is also shown on the plot. For a different piston configuration (2 horizontal pistons), the CN changed, but the system of equations is still well-conditioned, enabling recovery of the elasticities. This is shown in Fig. 8.

To show that inclusion size has no bearing on the reconstruction, an object with a small hard inclusion and a small soft inclusion was considered and the result of the reconstruction is shown in Fig. 3. With two horizontal pistons centered on the sides and one vertical piston centered on the top, the elasticity was reconstructed successfully using the computed displacements from the FP.

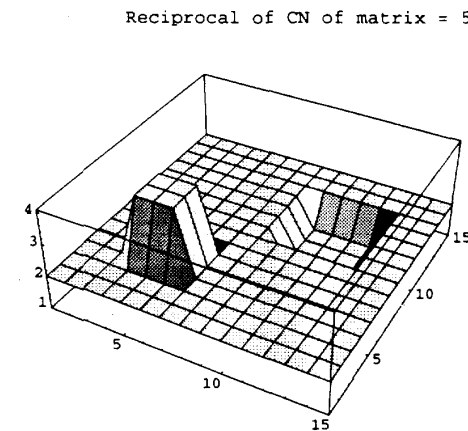


Figure 3: Reconstructed elasticity from solution to FP using 2 centered horizontal pistons and one centered vertical piston and object with both hard and soft inclusions.

The effect of noisy data was also considered. From the experiments, it appears that white Gaussian noise (WGN) is a good model for measurement noise. It was also seen in the experiments that the *absolute* noise level was about 1/75 of the level of the signal (size of deforming strain on the boundary). Using this noise model, WGN was added to the displacements² from the FP. A noisy estimate of the elasticity was obtained and is shown in Fig. 10, where both the hard and soft inclusions are still clearly seen.

V. CONCLUSION

We have developed a finite-difference based mathematical model to solve the forward and inverse problems of elasticity imaging. The IP equations $A'x' = b'$ are obtained by rearranging the FP equations $Ax = b$. The consistency of the forward and inverse problem procedures is significant since we intend to use them in a data-adaptive procedure

²Since the measurements are actually of the strains, the noise should be added to the strain. However since the strains are obtained from the displacements by differentiation, our simulations have a worse SNR than the experimental SNR.

in which the piston configuration is selected to best exhibit an inclusion (tumor), based on the estimate of the elasticity from the previous iteration. Simulation results for the FP agreed well with experimental results, and simulation results for the IP demonstrated that the approach is feasible.

One remaining problem is the effect of grid size on the CN of the matrix. As the grid size increases, the CN of the IP matrix becomes so large that the system is effectively singular due to ill-conditioning. We are currently investigating the use of regularized least squares methods to reconstruct the elasticity in these cases. Another issue to be considered is the availability of all the strain components, since one measured component of strain is more noisy than the other. We are considering the total least squares approach to ensure that noisy data in the system matrix A' does not produce an infeasible system $A'x = b'$.

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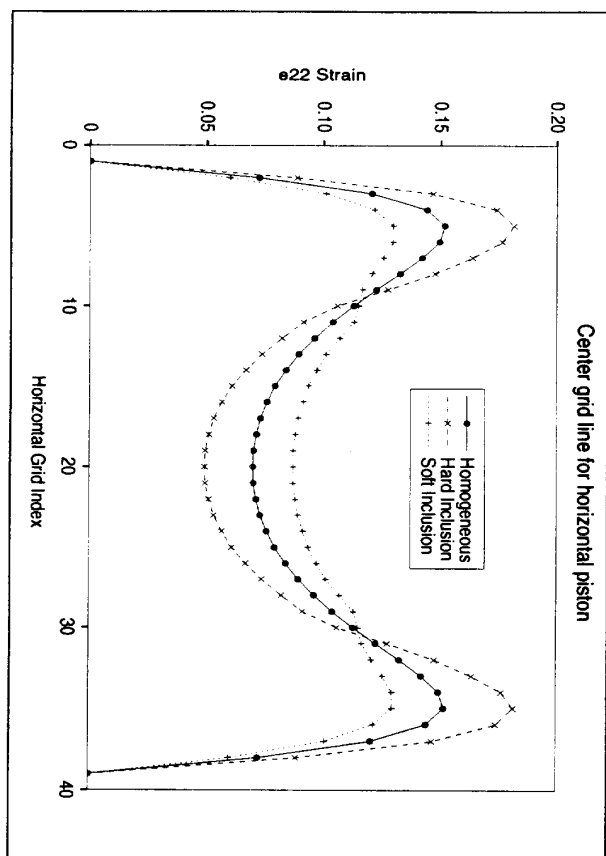


Figure 4: Horizontal center grid line ϵ_{22} component of the strain for pushes with 2 horizontal pistons centered on the sides of the object.

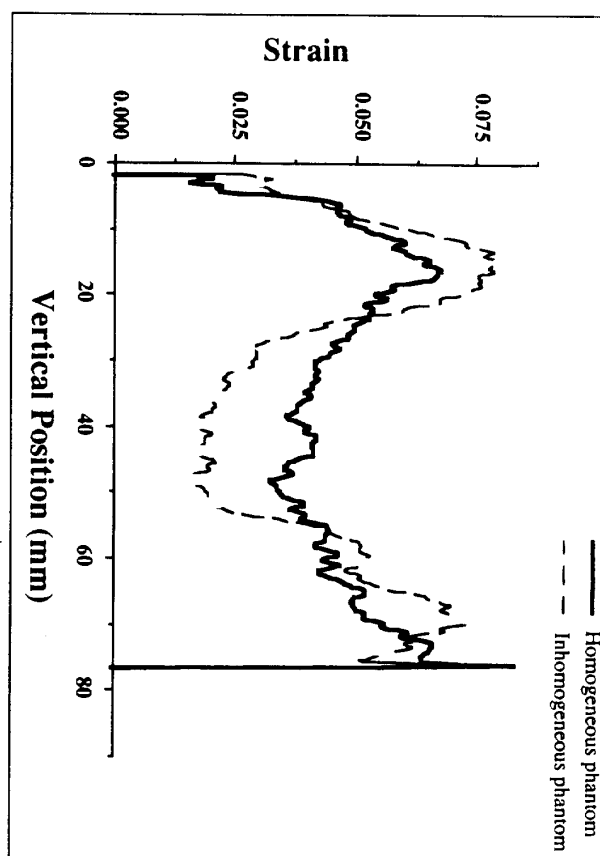


Figure 5: Experimentally measured vertical center grid line for ϵ_{22} component of the strain for push with a vertical piston centered on top of the object for a circular phantom.

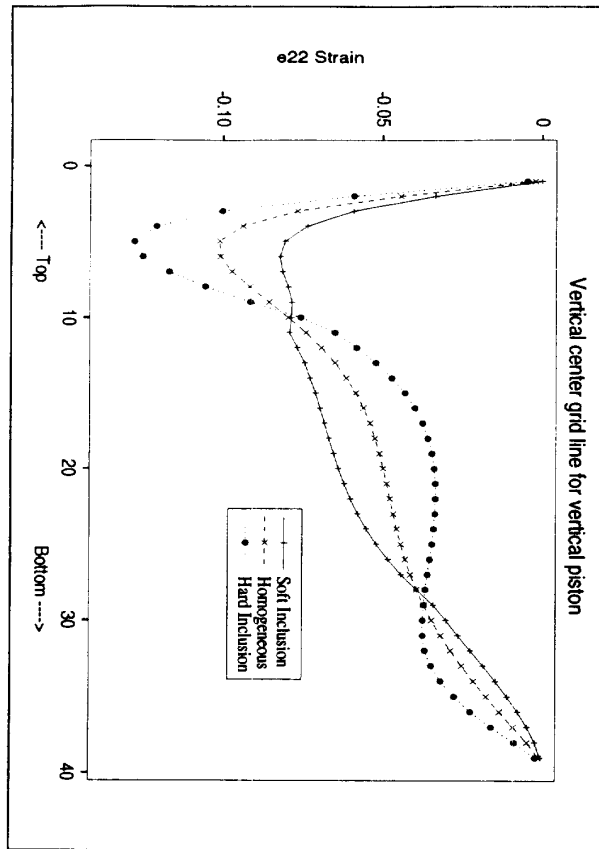


Figure 6: Vertical center grid line for ϵ_{22} component of the strain for push with a vertical piston centered on top of the object.

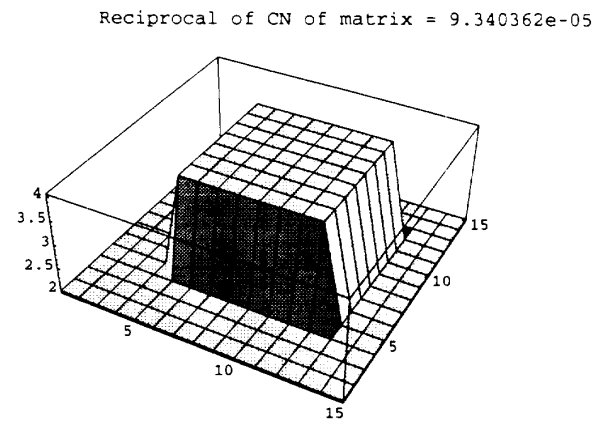


Figure 8: Reconstructed elasticity from solution to FP using 2 horizontal pistons centered on the sides and hard inclusion

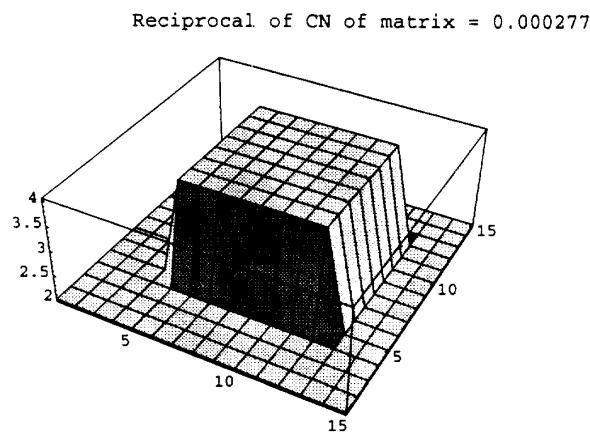


Figure 7: Reconstructed elasticity from solution to FP using vertical piston at center and hard inclusion

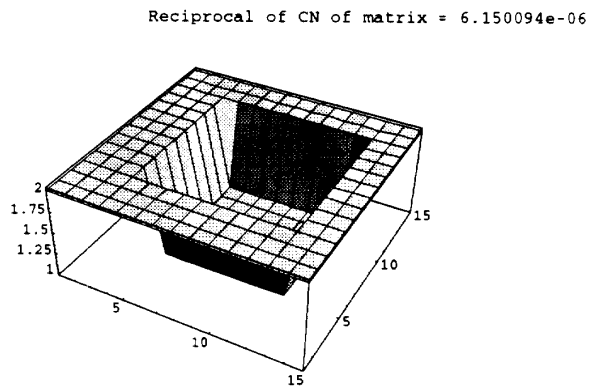


Figure 9: Reconstructed elasticity from solution to FP using 2 horizontal pistons centered on the sides and soft inclusion.

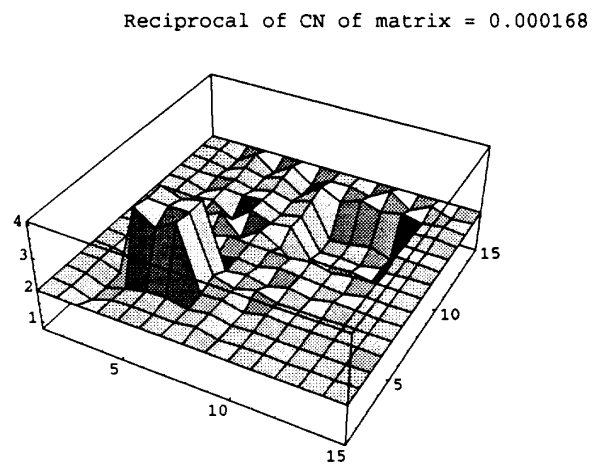


Figure 10: Reconstructed elasticity in the presence of noise.